

Quick Activity 1: School Supplies Calculator

A notebook costs **₱45**, and a pen cost **₱12**. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase

Input	Process	Output
Number of notebooks, number of pens, payment	Calculate notebook cost, calculate pen cost, add for total cost, calculate change	Total cost and change

Descomposition (Task)

1. Read the number of notebooks.
2. Read the number of pens.
3. Read the payment amount.
4. Calculate the notebook cost.
5. Calculate the pen cost.
6. Calculate the total cost.
7. Calculate the change.
8. Display the total cost and change.

Pseudocode

```
START
    READ notebooks, pens,
    payment
    notebookCost ←
    notebooks × 45
    penCost ←
    pens × 12
    totalCost ←
    notebookCost + penCost
    change ← payment - totalCost
    DISPLAY totalCost
    DISPLAY change
END
```

Dry Run Calculations

Test Case 1 Input:

- Notebook: 3
- Pen = 2
- Payments = ₱200

Step	Variable	Calculation	Solution
1	Notebook Cost	3×45	₱ 135
2	Pen Cost	2×12	₱ 24
3	Total Cost	$135 + 24$	₱ 159
4	Change	$200 - 159$	₱ 41

Output

Total Cost = ₱159

Change = ₱41

Test Case 2

Input:

- Notebook = 2
- Pens = 3
- Payment = ₱150

Step	Variable	Calculation	Solution
1	Notebook Cost	2×45	₱ 90
2	Pen Cost	3×12	₱ 36
3	Total Cost	$90 + 36$	₱ 126
4	Change	$150 - 126$	₱ 24

Output

Total Cost = ₱159

Change = ₱41

Final Answer

Test Case	Total Cost	Change
1	₱159	₱41
2	₱126	₱24

